

	Case Name: Recreation building	Sector	Construction (commercial building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Mayté & Thymo	Project Control	User defined distributions
Date	2019		Automatic tracking
File Name	C2019-09 Recreation building		Tracking based on user input

1. Project description

Project authenticity

A development team plans and constructs a fully equipped residential property, including site preparation, structural works, installations, and outdoor facilities, culminating in the delivery of a finished home with garden and amenities.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	45	Serial/Parallel (SP)	56%
Planned Duration (PD)	215	Activity Distribution (AD)	60%
Budget At Completion (BAC)	262,500 €	Length of Arcs (LA)	64%
Renewable Resources	-	Topological Float (TF)	8%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	100	0	N/A
CRI-tau	100	0	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	60	42	-0.36
SI	80	27	-1.43
SSI	10	12	1.15
CRI-r	14	11	1.07
CRI-rho	15	11	1.03
CRI-tau	10	7	0.81

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	50.23	-36.69	1	0	0
PV - SPI	38.94	24.23	CPI	0	0
PV - SCI	38.94	24.23	SPI	31.11	31.11
ED - 1	79.59	30.96	SPI(t)	29.79	29.79
ED - SPI	38.94	24.23	SCI	31.11	31.11
ED - SCI	38.94	24.23	SCI(t)	29.79	29.79
ES - 1	32.88	-32.79	0.8 CPI + 0.2 SPI	16.1	16.1
ES - SPI(t)	13.7	10.11	0.8 CPI + 0.2 SPI(t)	11.41	11.41
ES - SCI(t)	13.7	10.11			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-8500	-5000	-15.33	0.9	0.92	0.83	0.9
std dev	9539.39	0	12.5	0.06	0.03	0.14	0.03
final	-19500	-5000	-15	0.83	0.95	0.88	0.93

2.3.3.2. Time forecasting

PD	215	Real Duration	130	Late/early	39.53 %
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	219	0	1.86	-1.86
PV - SPI	235	8	9.3	-9.3
PV - SCI	260.67	12.58	21.24	-21.24
ED - 1	221.33	1.15	2.95	-2.95
ED - SPI	235	8	9.3	-9.3
ED - SCI	249	6.24	15.81	-15.81
ES - 1	230.33	12.5	7.13	-7.13
ES - SPI(t)	264.67	47.37	23.1	-23.1
ES - SCI(t)	281.67	46.76	31.01	-31.01

2.3.3.3. Cost forecasting

BAC	262,500 €	Real Cost	113,500.00 €	Over/under budget	56.76 %
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	276000	9539.39	143.17	-143.17
CPI	297286.07	22263.04	161.92	-161.92
SPI	295135.84	4799.71	160.03	-160.03
SPI(t)	323290.46	44581.19	184.84	-184.84
SCI	318079.45	15053.83	180.25	-180.25
SCI(t)	348634.74	44933.7	207.17	-207.17
0.8 CPI + 0.2 SPI	296600.55	17723.96	161.32	-161.32
0.8 CPI + 0.2 SPI(t)	301013.36	17747.61	165.21	-165.21